Solved Final Term Paper BT201 – 2018

QUESTIONS

BT201 – Ecology, Biodiversity and Evolution II

MCQs of BT201

1. Humans are included in the order....**Primates**
2. The field of biology specializing in storing and analyzing data... **bioinformatics**
3. The possibility of working with molecules on the scale of few micrometers is called.... **MICROECOLOGY**
4. Which one is **not** related to the human genome project... ***other than following three***

• synteny analysis, • phylogenetic footprinting, • phylogenetic shadowing

5.First question asked when a species is bought for genome sequencing... **what is the size of its genome?**
6.The study of transcripts that are present at any time in the cell are called...**transcriptomics**
7.Important measures used to describe community composition are.... **species richness and species diversity**
8.Vertebrates makes....%of the animal…**5%**
9.the number of allele in a sample.... **Allelic richness**
10.Which is not the part of horizontal gene transfer... **Conjugation • Transformation • Transduction**
11.mass spectrometry used to determine....sequence ,,, **amino acid sequence of a protein**
12. Oparin (1924): work… "**Primeval soup" theory**

3)Quinine is obtained from quina quina it is used to treat **malaria**
6)which is not related protozoa reproduce by **parthenogenesis**….Answer
7)the common extension of genomics are except Answer **Vaccinomics**
2. To assemble a genome from a series of sequence fragments requires tremendous
*computational power
3. The first genome sequenced completely was that of
*Hemofelues influenza
4. The ---------- technique has now been replaced mostly by sequencing using mass spectrometry.
*Edman degradation
5. Species differ from foundation species in that they are more like to belong to ***higher trophic level, diverse action, modify environment** ***(All above)***
6. Annelids do not have feeding style ***filter feeders**
8. Modern bread wheat is a mixture of recombined genes from **b) 3**
14. Gymnosperms means **Naked seeds**
16. Medicines from nature account for usage by ---
-------- of the world population. **80 per**cent
17. Habitat destruction is the major cause for
Exact wording is Biotdiversity loss.
19. Coral reefs which are biodiversity hotspots will disappear in
20 to 40 years

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22. Plants found in a particular area is **a. Flora**

21. Gregor mendel experiments with plant hybridization led to this establishing of Ans is "Laws of inheritance"

24. Genetic modification of plant is achieved by adding a specific gene or genes to a plant or by knocking down a gene with to produce a desirable phenotype. a. RNAi

25. Which organization is not involved in human genome project….??

26. they posses notochord in any period of their life… **Chordata**

Define community structure? 2 marks

Community structure is essentially the composition of a community, including the number of species in that community and their relative numbers. It can also be interpreted more broadly, to include all of the patterns of interaction between these different Species.

what is intermediate distribution hypothesis. (2)

The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

Ecological genomics..2

A scientific discipline that studies the structure and functioning of a genome with the aim of understanding the relationship between the organism and its biotic and abiotic environments

Enlist four reasons for lacking of fossil record?2 marks

1. Soft tissues are rarely preserved
2. Movement of the earth's crust has obliterated and/or covered many fossils
3. Fossilization takes place only in certain types of habitats and favorable environments
4. Paleontologists have not dug up every place on earth

Explain species richness and species diversity with example? 3 marks

Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first. Communities with the highest species richness tend to be found in areas near the equator, which have lots of solar energy (supporting high primary productivity), warm temperatures, large amounts of rainfall, and little seasonal change. Communities with the lowest species richness lie near the poles, which get less solar energy and are colder, drier, and less amenable to life

Species diversity is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness). Larger numbers of species and more even abundances of species lead to higher species diversity. For example: • A forest community with 20 different kinds of trees would have greater species diversity than a forest community with only 5 kinds of trees (assuming that the tree species were even in abundance in both cases)

Explain importance of biodiversity? 5 marks These Are As Follows:

a. Maintaining Balance of The Ecosystem.
b. Recycling And Storage of Nutrients,
c. Combating Pollution, And Stabilizing Climate, Protecting Water Resources,
d. Forming And Protecting Soil And Maintaining Eco-Balance.
e. Intrinsic Value
f. A Philosophical Concept of The Value of Something Independent of Its Value To Anyone or Anything

What are phylogenetic trees (How genes are related to each other) ? 5 marks

Explore genome evolution based on large data sets of DNA or protein sequences. Using entire genomes to infer a species tree (Eisen and Fraser 2003). Based on maximum gene

Factors affacting comunity structure

a. The climate patterns of the community's location.
b. The geography of the community's location
c. The heterogeneity (patchiness) of the environment
d. The frequency of disturbances, or disruptive events
e. Interactions between organisms

Enlist three major technological developments of 1990s.

MICROECOLOGY • COMPUTING • COMMUNICATION.

Enlist types of servers used in bioinformatics.

WWW servers
Database servers
Intensive computing servers

How sequence assembly is done using shotgun approach?

DNA is broken up randomly into numerous small segments, which are sequenced using the chain termination method to obtain reads. Multiple overlapping reads for the target DNA are obtained by performing several rounds of this fragmentation and sequencing. Computer programs then use the overlapping ends of different reads to assemble them into a continuous sequence.[

What phyla include amoeboid protozoan..Write their mode of locomotion...

Amoebozoa

It is a crawling-like type of movement accomplished by protrusion of cytoplasm of the cell involving the formation of pseudopodia

How the evolutionary history convert in to document?

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- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

Write at least three reasons to use phylogenetic.

Understand the lineage of different species
Organizing principle to sort species into taxonomy
Understand how various functions evolved
Understand forces and constraints on evolution
Perform multiple sequence alignment
Predict gene function (phylogenetic footprint)

Differentiate between species tree and gene tree.

Species tree (how are my species related?) contains only one representative from each species when did speciation take place? all nodes indicate speciation events **Gene tree** (how are my genes related?) often contains a number of genes from a single species. nodes relate either to speciation or gene duplication events.

Define genome

A genome is an organism's complete set of DNA, including all of its genes. Each genome contains all of the information needed to build and maintain that organism.

Explain Sequencing?? 3

How genes are related to other genes? ..5